

Random Variables & Probability Distributions

Statistics Worksheet · Grade 9–12

Name: _____

Date: _____

Learning Objectives

- Define random variables and identify their possible values from a given experiment
- Distinguish between discrete and continuous random variables
- Construct a probability distribution table and calculate probabilities for discrete random variables

Problems

1. A fair coin is flipped once. Let X be the number of heads. List all possible values of the random variable X .

$$X = ?$$

2. A single fair die is rolled. Let X be the number showing on top. List all possible values of the random variable X .

$$X = ?$$

3. A coin is flipped three times. Let X be the number of tails obtained. List all possible values of the random variable X and state how many total outcomes are in the sample space.

$$X = ?, \quad |S| = ?$$

4. Two dice are rolled. Let X be the sum of the two numbers facing up. What are the minimum and maximum possible values of X ?

$$X_{\min} = ?, \quad X_{\max} = ?$$

5. A coin is flipped four times. Let X be the number of tails. Complete the probability distribution table below by finding the probability for each value of X . Use the fact that the total sample space has 16 outcomes.



X (# of Tails)	Number of Ways	P(X)
0	1	
1	4	
2	6	
3	4	
4	1	

6. Using the probability distribution from the four-coin-flip experiment (flipping a coin four times, X = number of tails), find the probability of getting at least 3 tails.

$$P(X \geq 3) = ?$$

7. A bag contains 3 red marbles and 2 blue marbles. Two marbles are drawn one at a time without replacement. Let X be the number of red marbles drawn. List all possible values of X and find $P(X = 2)$.

$$P(X = 2) = ?$$

8. Using the same bag from Problem 7 (3 red, 2 blue marbles, drawing 2 without replacement), complete the full probability distribution table for X = number of red marbles drawn.

X	P(X)
0	
1	
2	

9. A discrete random variable X has the probability distribution shown below. Find the missing probability $P(X = 3)$ and then calculate the expected value (mean) of X .

X	P(X)
1	0.10
2	0.25
3	



X	P(X)
4	0.20
5	0.15

10. Two dice are rolled. Let X be the sum of the two numbers facing up. Find $P(X = 7)$ and $P(X \text{ is even})$. Express each as a fraction in simplest form. (Hint: There are 36 equally likely outcomes when rolling two dice.)

$P(X = 7) = ?$, $P(X \text{ is even}) = ?$



Random Variables & Probability Distributions — Answer Key

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Answer Key

1. Answer: $X = 0, 1$

- When flipping a coin once, the outcomes are Heads (H) or Tails (T).
- If X counts the number of heads: getting T gives $X = 0$, getting H gives $X = 1$.
- Therefore, $X = \{0, 1\}$.

2. Answer: $X = 1, 2, 3, 4, 5, 6$

- A standard die has faces numbered 1 through 6.
- Each face is a possible outcome, so X can take any value from 1 to 6.
- Therefore, $X = \{1, 2, 3, 4, 5, 6\}$.

3. Answer: $X = 0, 1, 2, 3; |S| = 8$

- When flipping a coin 3 times, the total number of outcomes is $2^3 = 8$.
- The number of tails can range from 0 (all heads: HHH) to 3 (all tails: TTT).
- Therefore, $X = \{0, 1, 2, 3\}$ and $|S| = 8$.

4. Answer: $X_{\min} = 2, X_{\max} = 12$

- The smallest number on each die is 1, so the minimum sum is $1 + 1 = 2$.
- The largest number on each die is 6, so the maximum sum is $6 + 6 = 12$.
- Therefore, X ranges from 2 to 12.

5. Answer: See completed table

X (# of Tails)	Number of Ways	$P(X)$
0	1	$1/16$
1	4	$4/16 = 1/4$
2	6	$6/16 = 3/8$
3	4	$4/16 = 1/4$
4	1	$1/16$

- Total sample space = $2^4 = 16$ outcomes.
- $P(X = 0) = 1/16$ (only HHHH has 0 tails).
- $P(X = 1) = 4/16 = 1/4$ (four ways to get exactly 1 tail).
- $P(X = 2) = 6/16 = 3/8$ (six ways to get exactly 2 tails).
- $P(X = 3) = 4/16 = 1/4$ (four ways to get exactly 3 tails).



- Check: $1 + 4 + 6 + 4 + 1 = 16$, and all probabilities sum to 1.

6. Answer: $P(X \geq 3) = 5/16$

- $P(X \geq 3)$ means $P(X = 3) + P(X = 4)$.
- $P(X = 3) = 4/16$ and $P(X = 4) = 1/16$.
- $P(X \geq 3) = 4/16 + 1/16 = 5/16$.

7. Answer: $X = 0, 1, 2$; $P(X = 2) = 3/10$

- X can be 0, 1, or 2 (the number of red marbles drawn out of 2).
- Total ways to choose 2 from 5 marbles: $C(5,2) = 10$.
- Ways to choose 2 red from 3 red: $C(3,2) = 3$.
- $P(X = 2) = 3/10$.

8. Answer: See completed table

X	$P(X)$
0	1/10
1	$6/10 = 3/5$
2	3/10

- Total ways to pick 2 marbles from 5: $C(5,2) = 10$.
- $P(X = 0)$: Choose 0 red and 2 blue: $C(3,0) \times C(2,2) = 1 \times 1 = 1$. $P = 1/10$.
- $P(X = 1)$: Choose 1 red and 1 blue: $C(3,1) \times C(2,1) = 3 \times 2 = 6$. $P = 6/10 = 3/5$.
- $P(X = 2)$: Choose 2 red and 0 blue: $C(3,2) \times C(2,0) = 3 \times 1 = 3$. $P = 3/10$.
- Check: $1/10 + 6/10 + 3/10 = 10/10 = 1$. ✓

9. Answer: $P(X = 3) = 0.30$; $E(X) = 3.05$

- All probabilities must sum to 1: $0.10 + 0.25 + P(X=3) + 0.20 + 0.15 = 1$.
- $0.70 + P(X=3) = 1$, so $P(X=3) = 0.30$.
- $E(X) = 1(0.10) + 2(0.25) + 3(0.30) + 4(0.20) + 5(0.15)$.
- $E(X) = 0.10 + 0.50 + 0.90 + 0.80 + 0.75 = 3.05$.

10. Answer: $P(X = 7) = 6/36 = 1/6$; $P(X \text{ is even}) = 18/36 = 1/2$

- Total outcomes when rolling two dice: $6 \times 6 = 36$.
- Ways to get a sum of 7: (1,6),(2,5),(3,4),(4,3),(5,2),(6,1) → 6 ways. $P(X=7) = 6/36 = 1/6$.
- Even sums possible: 2, 4, 6, 8, 10, 12.
- Count outcomes for each: sum=2→1, sum=4→3, sum=6→5, sum=8→5, sum=10→3, sum=12→1. Total = 18.
- $P(X \text{ is even}) = 18/36 = 1/2$.

